

DILEEP SIVARAMAN

Present Affiliation:

Foreign Expert,
Center for Biomedical and Robotics Technology
(BART LAB),
Faculty of Engineering,
Mahidol University,
Salaya, Thailand 73170

Email ID: dileepsivaraman89@gmail.com
: dileep.siv@mahidol.ac.th

Phone: +66843410548

Google Scholar: https://scholar.google.com/citations?user=R_I4Ut0AAAAJ&hl=en&oi=ao

Website:

<https://dileepsivaraman007.github.io/dileepsivaraman/>

HOME:

Dileep Bhavanam
Kadampanad South PO
Pathanamthitta (DST)
Kerala,
Pin- 691553
India



RESEARCH PROFILE

I am a researcher working at the intersection of robotics, biomedical engineering, and physical system modelling, with a focus on bio-inspired systems, system identification, and probabilistic state estimation. My work centres on sensorless force estimation and the modelling of dynamical interactions in medical robotics, with growing interest in exploiting the dynamics of physical systems for sensing, estimation, and control.

RESEARCH INTERESTS IN KEYWORDS

- **Robotics and Bio-Inspired Systems:** Bio-inspired robotic mechanisms, embedded sensing and force estimation, smart material, medical robotics, robotic system modeling
- **Estimation and Signal Processing for Physical Systems:** Kalman filtering, adaptive polynomial filtering, sensor fusion, nonlinear state estimation, time-series modeling
- **Control and Learning for Dynamical Systems:** System identification, state estimation, adaptive filtering, uncertainty quantification

EDUCATION

- | | |
|--------------|---|
| Feb 17, 2025 | Ph.D. Biomedical Engineering (Medical Robotics) - Mahidol University, Thailand |
| 2012 | Integrated M.Sc. (Physics & Mathematics) - Amrita Vishwa Vidyapeetham University, Amritapuri Campus, Kollam, Kerala, India |

RESEARCH EXPERIENCE

Foreign Expert – BART LAB, Mahidol University (2025–Present)

Research focus: data-driven estimation and modeling of robotic and bio-inspired systems.

PhD Researcher – Mahidol University (2020–2025)

Topic: Developed adaptive state estimation frameworks for bio-inspired needle systems in minimally invasive surgical robotics, with application to breast biopsy.

SCHOLARLY PUBLICATIONS

International Journals

- Sivaraman, Dileep, et al. "Dynamic order selection analysis in adaptive polynomial Kalman filtering: implementation and integration of sensor data and hybrid image processing for bio-inspired needle systems." *Systems Science & Control Engineering* 13.1 (2025): 2546839.

- Kunapinun, Alisa, et al. **"Toward AI-Powered Neurovascular Intervention: From Imaging to XR-Robotic Convergence."** *Stroke*.
- Sivaraman, D., Pillai, B. M., Ongwattanakul, S., & Suthakorn, J. **"Adaptive Polynomial Predictive Filter for Sensor Data Estimation and Prediction in Interference Environment."** *IET Control Theory and Applications* (2024).
- Sivaraman, D., Ongwattanakul, S., Moonjaita, C., Suthakorn, J., & Madhavan Pillai, B. **"A pack hunting strategy for heterogeneous robots in rescue operations.."** *Bioinspiration & Biomimetics* (2024).
- Sivaraman, Dileep, Songpol Ongwattanakul, Branesh M. Pillai, and Jackrit Suthakorn. **"Adaptive polynomial Kalman filter for nonlinear state estimation in modified AR time series with fixed coefficients."** *IET Control Theory & Applications* (2024).
- Pillai, Branesh M., Jackrit Suthakorn, Dileep Sivaraman, Sakol Nakdhamabhorn, Nantida Nillahoot, Songpol Ongwattanakul, Fumitoshi Matsuno, Mikhail Svinin, and Evgeni Magid. **"A heterogeneous robots collaboration for safety, security, and rescue robotics: e-ASIA joint research program for disaster risk and reduction management."** *Advanced Robotics* 38, no. 3 (2024): 129-151.
- Pillai, Branesh M., Peerapat Owatchaiyapong, Shen Treratanakulchai, Dileep Sivaraman, Songpol Ongwattanakul, and Jackrit Suthakorn. **"Lower Limb Exoskeleton with Energy-Storing Mechanism for Spinal Cord Injury Rehabilitation."** *IEEE Access* 11 (2023): 133850-133866.
- Sivaraman, Dileep, Songpol Ongwattanakul, Jackrit Suthakorn, and Branesh M. Pillai. **"Nonlinear Dynamic States' Estimation and Prediction Using Polynomial Predictive Modeling.."** *IEEE Canadian Journal of Electrical and Computer Engineering* 46, no. 3 (2023): 185-195.

International Proceedings

- Moonjaita, Choladawan, et al. **"Ground Reaction Force Analysis of Robot-Assisted Walking Using Inertial Sensors."** *2025 18th International Convention on Rehabilitation Engineering and Assistive Technology (i-CREATE)*. IEEE, 2025.
- Sivaraman, D., Wiratkapun, C., Pillai, B. M., Suthakorn, J., & Ongwattanakul, S. **"Shape Memory Alloy Spring-Based Force Sensing System for Dual-Sheath Surgical Needle."** *2025 IEEE International Conference on Robotics and Biomimetics (IEEE ROBIO)*. IEEE, Chengdu, China.
- Sivaraman, D., Wiratkapun, C., Pillai, B. M., Suthakorn, J., Methachan, B., Nakdhamabhorn, S., & Ongwattanakul, S. **"Hybrid image-based motion tracking and adaptive polynomial Kalman filter for a bio-inspired dual-sheath needle system."** *2024 IEEE International Conference on Robotics and Biomimetics (IEEE ROBIO)*. IEEE, Bangkok, Thailand.
- Pillai, B. M., Siripala, N., Sai-aroon, K., Sivaraman, D., Vinjamuri, R., Chumnavej, S., & Suthakorn, J. **"Enhancing post-stroke upper limb rehabilitation through haptic feedback in virtual reality-based gaming."** *2024 IEEE International Conference on Robotics and Biomimetics (IEEE ROBIO)*. IEEE, Bangkok, Thailand.
- Sivaraman, D., et al. **"Adaptive Polynomial Predictive Filter: Solving Inconsistent and Interrupted Sensor Data Challenges."** *2023 IEEE International Conference on Robotics and Biomimetics (ROBIO)*. Ko Samui, Thailand.
- Sivaraman, D., et al. **"Energy Optimized Path Planning and Decision Making for Multiple Robots in Rescue Operations."** *IECON 2022–48th Annual Conference of the IEEE Industrial Electronics Society* (pp. 1-6). IEEE, Brussels, Belgium, 2022.
- Sivaraman, D., et al. **"Sensorless Based Gravity Torque Estimation and Friction Compensation for Surgical Robotic System."** *2022 IEEE 9th International Conference on e-Learning in Industrial Electronics (ICELIE)* pp. 1-6. IEEE, Brussels, Belgium, 2022.
- Sivaraman, D., et al. **"Energy Optimized Path Planning and Decision Making for Multiple Robots in Rough Terrain."** *The Twenty-Seventh International Symposium on Artificial Life and Robotics 2022 (AROB 27th 2022)*, *The Seventh International Symposium on BioComplexity 2022 (ISBC 7th 2022)*, *The Fifth International Symposium on Swarm Behavior and Bio-Inspired Robotics 2022 (SWARM 5th 2022)*, ONLINE, January 25-27, 2022.

Peer-Reviewed Local Proceedings

- Sivaraman, D., et al. **"Curvature Estimation of Steerable Flexible Probe for Ductal Carcinoma in Situ Detection: A Simulation Study."** *18th Asian Conference on Computer Aided Surgery (ACCAS 2022) and 5th International Conference on Business, Informatics, and Management (ICBIM 2022)*, Aug 2022.

Book Chapter

- Sivaraman, D., Pillai, B. M., Suthakorn, J., & Ongwattanakul, S. (2024). **Human–Robot Interaction in Biopsy Procedures: A Biomimetic Dual-Sheath Needle Design Inspired by Insect Ovipositor Mechanics.** In *Discovering the Frontiers of Human-Robot Interaction: Insights and Innovations in Collaboration, Communication, and Control* (pp. 401-417). Springer Nature Switzerland.

Manuscripts Under Review / Submitted

- Dileep Sivaraman, Branesh M. Pillai, Jackrit Suthakorn, Asoke K. Nandi, and Songpol Ongwattanakul, **Adaptive Polynomial Kalman Filtering and Model Order Selection: Comparative Analysis with Polynomial and Autoregressive Representations.** (*Digital Signal Processing: A Review Journal*, Under Review)
- Dileep Sivaraman, Branesh M. Pillai, Jackrit Suthakorn, **Error-Dependent Bandwidth Shaping for DC Motor Control via Analytical Pole Placement.** (*Mechatronics*, Submitted)
- Dileep Sivaraman, Branesh M. Pillai, and Jackrit Suthakorn, **Direction-Dependent Friction Identification and Compensation in Brushed DC Servo Motors Using Closed-Loop Disturbance Observer Data.** (*IEEE Transactions on Industrial Electronics*, Submitted)

FELLOWSHIPS, GRANTS, AND ACADEMIC SUPPORT

- **Travel Grant**, Faculty of Engineering, Mahidol University, to attend and present at the IEEE International Conference on Robotics and Biomimetics (ROBIO 2025), Chengdu, China [Dec 3–7, 2025].
- **The Reinventing University System has funded the research**, supported through Mahidol University (IO 864102063000), and Research Funding by Mahidol University under the project: Integrating Navigation and Robotics for breast biopsy using Imaging and Models (Pre-Clinic and Clinic Evaluation Phase for Product Commercialization) [2023-24]
- **Mahidol Postgraduate Scholarships** for Graduate International Students [2020-2023].

ACADEMIC DISTINCTIONS AND SCHOLARLY ACTIVITIES

- Participated in RoboCup 2023 held in Bordeaux, France, from July 4th to 10th, 2023, as a member of the BART LAB ROBOTICS Team.

PROFESSIONAL ACTIVITIES – ACADEMIA (CONFERENCES)

- Section Chair, 2025 IEEE International Conference on Robotics and Biomimetics (IEEE ROBIO).
- Academic Committee Member, 18th International Convention on Rehabilitation Engineering and Assistive Technology (i-CREATE 2025), November 2025
- IOP Peer Review Excellence Certificate - (2025) Institute of Physics Publishing

RESEARCH PROJECTS AND CONTRIBUTIONS

Faculty of Engineering, Mahidol University

- **Task-Oriented Link Length Optimization for Mobile Manipulators in Disaster Response (2026–Present)**
 - Conducting kinematic design and workspace optimization of mobile manipulators for multi-task disaster response scenarios
 - Applied the framework to RoboCup Rescue–style manipulation tasks, emphasizing real-world deployability
- **Advanced Research in Adaptive Polynomial Kalman Filtering and Optimization (2023–Present)**
 - Collaborative research project with Prof. Asoke Nandi, Brunel University London (UK)

- Focused on adaptive polynomial Kalman filters (APKF) for improved nonlinear state estimation in time-varying systems
- Developed and analyzed model order selection techniques, μ -term augmentation, and hybrid AR-polynomial structures
- **Nonlinear hysteresis-aware state estimation for SMA-based minimally sensed surgical systems (2025–Present)**
 - Developing an SMA-based embedded force sensing mechanism for minimally invasive Bio-inspired needle systems
 - Investigating nonlinear electromechanical and hysteretic behavior of SMA elements under constrained insertion dynamics
 - Formulating a physics-constrained state estimation framework for real-time internal force reconstruction under limited sensing
- **Bio-inspired needle for Breast Biopsy Using a Novel State Estimation/Prediction Algorithm (2021–2025)**
 - Supervised by Dr. Songpol Ongwattanakul, Dr. Jackrit Suthakorn (Co-Advisor, BART LAB) and Prof. Cholatip Wiratkapun (Co-Advisor) clinician from Department of Radiology, Faculty of Medicine Ramathibodi Hospital, Mahidol University, serving as co-advisor.
 - Conducted interdisciplinary research integrating robotic system design with clinical requirements for biopsy procedures
- **e-ASIA Joint Research Program (e-ASIA JRP) Disaster Risk and Reduction Management (Torrential Rainfall, Associated Flood and Land Slide) (2022-2023)**
 - A multidisciplinary international collaboration with Kazan Federal University (Russia) and Kyoto University (Japan)
 - Focused on robotic solutions and AI-driven strategies for disaster response
 - Contributed to research on unmanned systems, terrain modeling, and sensor integration in harsh environments
- **Development of Unmanned Aerial Vehicles for Disaster Risk and Reduction Management (2022-2023)**
 - Developed a bioinspired path planning strategy for heterogeneous robotic systems operating in uncertain terrain.
- **Stairclimbing Wheelchair with Navigation and Mapping. (2022)**
 - Assisted in the assembly, integration, and testing of mechanical components.

Interdisciplinary and Applied Collaborations

- **Mask Material Filtration Efficiency Test - Along with Faculty of Medicine, Siriraj Hospital, Mahidol University. (2020)**
 - Conducted testing in coordination with medical professionals, ensuring compliance with hospital-grade standards.

MASTER'S THESIS

Title: Photometric studies of galactic open star cluster Berkeley 7

Supervisor: Dr. Sneha Lata, Aryabhata Research Institute of Observational Sciences

(Department of Science and Technology, Government of India), Manora Peak, Nainital, India

PROJECT AT SCHOOL OF ARTS SCIENCE, AMRITA VISWA VIDYAPEEDAM UNIVERSITY

Title: Dynamics of Nonlinear Maps

Supervisor: Dr. V. M Nandakumaran, Head of the Department of Physics

Amrita Vishwa Vidyapeetham University, Amritapuri Campus, Kerala, India

TEACHING AND MENTORSHIP

Foreign Expert Teaching

SI No	Name of the Subject	Year	Level	Role
1	EGBI481: Introduction to Medical Robotics	2025	UG(BME)	Shared with Prof. Jackrit Suthakorn
2	EGBE 680: Advanced Medical Robotics	2026	PG(BME)	Shared with Prof. Jackrit Suthakorn

Adjacent Faculty - Asian institute of Technology- Thailand

SI No	Name of the Subject	Year	Level	Role
1	Self-Driving Vehicles in Smart City	2026	PG	Shared with Dr.Branesh

Doctoral Teaching Assistantship

SI No	Name of the Subject	Year	Level	Role
1	Computational Methods for Biomedical Engineering (Python)	2023-2024	UG (BME)	Shared with Dr. Songpol Ongwattanakul
2	Healthcare Automation	2024	PG (BME)	Shared with Dr. Songpol Ongwattanakul

Academic and Professional Appointments

- Lecturer at Department of Physics, Shree Vidyadhiraja College of Arts & Science, Karunagappally, Kerala University, Kerala, India (2017-2019)
- National Service Scheme - Programme Officer at Shree Vidyadhiraja College of Arts & Science, Karunagappally, Kerala University, Kerala, India (2017-2019)
- Lecturer at Al-Azhar Group of Institutions, Thodupuzha, Department of Technical Education (DTET) of Kerala, Kerala, India (2013-2017)

SCHOLARLY TRAINING AND WORKSHOPS

- Workshop on '**Recent Trends in Physics**' 2011 was organized by the Indian Academy of Sciences, National Academy of Sciences, and Indian National Science Academy.

PROFESSIONAL ACTIVITIES – INVITED SPEAKER/KEYNOTE SPEAKER

- **Keynote Speaker** – 1st International Conference on AI Innovations in Medical Engineering and Robotics Technology (AI-MERT 2025), to be held at the Asian Institute of Technology (AIT), Thailand, on December 18–19, 2025.

COMPUTER SKILLS

- Programming: MATLAB, Python, and ROS
- Mechanical Designing: SolidWorks, Onshape
- Web Development: GitHub Pages, Google Sites, HTML/CSS, Git version control
- Content Writing: LaTeX for academic publications, technical documentation, and research reports

ACADEMIC AND COMMUNITY ENGAGEMENT

- Technical Volunteer - RoboCup 2022, Bangkok

– Volunteer (Flood relief)- Plan International (India Chapter) 2018, Kerala, India

REFERENCE CONTACTS

Dr. Songpol Ongwattanakul

Executive Director, Center for Biomedical and Robotics Technology, (BART LAB), Department of Biomedical Engineering, Faculty of Engineering, Mahidol University, Salaya, Thailand 73170

Email: songpol.ong@mahidol.ac.th | <https://bartlab.org/>

Dr. Branesh Madhavan Pillai

Assistant Professor, Department of Industrial Systems Engineering (ISE), School of Engineering and Technology (SET), Asian Institute of Technology (AIT), Pathum Thani 12120 Thailand

Email: branesh@ait.asia | www.ait.ac.th

Dr. Ramana Vinjamuri

Associate Professor of Computer Science and Electrical Engineering, University of Maryland, Baltimore County, 1000 Hilltop Circle, Baltimore, MD 21250

Email: rvinjam1@umbc.edu